

SSG-LUGIA: Single Sequence based Genome Level Unsupervised Genomic Island Prediction Algorithm

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Abstract

Background: Genomic Islands (GIs) are clusters of genes that are mobilized through horizontal gene transfer. GIs play a pivotal role in bacterial evolution as a mechanism of diversification and adaptation to different niches. Therefore, identification and characterization of GIs in bacterial genomes is important for understanding bacterial evolution. However, quantifying GIs is inherently difficult, and the existing methods suffer from low prediction accuracy and precision–recall trade-off. Moreover, several of them are supervised in nature, and thus, their applications to newly sequenced genomes are riddled with their dependency on the functional annotation of existing genomes.

Results: We present SSG-LUGIA, a completely automated and unsupervised approach for identifying GIs and horizontally transferred genes. SSG-LUGIA is a novel method based on unsupervised anomaly detection technique, accompanied by further refinement using cues from signal processing literature. SSG-LUGIA leverages the atypical compositional biases of the alien genes to localize GIs in prokaryotic genomes. SSG-LUGIA was assessed on a large benchmark dataset ‘IslandPick’

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and on a set of 15 well-studied genomes in the literature and followed by a thorough analysis on the well-understood *Salmonella typhi* CT18 genome. Furthermore, the efficacy of SSG-LUGIA in identifying horizontally transferred genes was evaluated on two additional bacterial genomes, namely, those of *Corynebacterium diphtheria* NCTC13129 and *Pseudomonas aeruginosa* LESB58. SSG-LUGIA was examined on draft genomes and was demonstrated to be efficient as an ensemble method.

Conclusions: Our results indicate that SSG-LUGIA achieved superior performance in comparison to frequently used existing methods. Importantly, it yielded a better trade-off between precision and recall than the existing methods. Its nondependency on the functional annotation of genomes makes it suitable for analyzing newly sequenced, yet uncharacterized genomes. Thus, our study is a significant advance in identification of GIs and horizontally transferred genes. SSG-LUGIA is available as an open source software at <https://nibtehaz.github.io/SSG-LUGIA/>.

Key words: Genomic Islands; Pathogenicity Islands; horizontal gene transfer; unsupervised learning; anomaly detection

Introduction

Bacteria, the most prevalent component of biomass [1], exhibit an astounding amount of diversity in metabolic and cellular structures and functionalities [2]. During 1990s, virulence genes were discovered in some strains of *Escherichia coli* in a stark contrast to nonpathogenic *E. coli* strains [3]. Clusters of virulence genes in the pathogenic *E. coli* strains were thought to be the result of horizontal gene transfer (HGT) events. HGT, also referred to as lateral gene transfer (LGT), refers to the exchange of genetic material among organisms in a way other than the vertical transmission of genes from parents to offsprings through reproduction [4–6]. HGT plays an important role in the evolution, diversification and adaptation of bacteria. Such events are believed to be responsible for diverse features of bacteria including their resistance to antibiotics, ability to degrade artificially synthesized substances, flourish in novel environments or to pulverize the threats of decimation or even extinction when faced with natural changes or targeted human efforts [7].

The gene clusters acquired via HGT and harboring virulence-associated genes were termed as pathogenicity islands (PAIs) [8]. Later studies, however, revealed that horizontally acquired gene clusters were capable of performing other tasks as well. This led to the discovery of different types of islands, such as secretion islands, antimicrobial resistance islands and metabolic islands [9]. Later on, these clusters of genes of various functionalities, ranging from 10 to 200 Kbp in length and acquired via HGT, were collectively termed as Genomic Islands (GIs) [10].

Realization of the significance and impact of HGT on bacterial evolution has led to the development of a number of methods to identify GIs as well as laterally acquired genes in bacterial genomes. These methods can be classified mainly into two types, composition-based methods and comparative genomics methods. Composition-based methods usually work on a single-genome sequence and predict GIs or ‘alien’ genes based on the atypical characteristics of the laterally transferred regions. On the contrary, comparative genomics-based approaches identify clusters of alien regions by comparing a number of closely related genomes, by analyzing which genes or genomic regions are present in one genome but not in the others. Tools like BLAST [11], Mauve [12] and MUMmer [13] are used for the latter type of approaches. These methods, however, require multiple genomes of closely related organisms. In addition, phylogenetic tree-based methods are also invoked frequently to catalog alien genes in prokaryotic or eukaryotic genomes [14].

Composition-based methods are often more desirable as they require only a single genome; however, they suffer from numerous false positives and false negatives, partly due to the inability

of these methods to detect transfers among compositionally similar but phylogenetically distinct organisms, as well as the inability to screen native genes that are compositionally deviant for reasons other than HGT [8]. SIGI-HMM predicts GIs based on codon usage bias [15], which has a reasonable precision but low recall [14]. PAI-IDA works with the GC content as well as dinucleotide and codon usage [16]; in contrast, IslandPath-DIMOB not only measures dinucleotide biases but also takes into account the presence of mobility genes [17–19]. Alien Hunter predicts HGT events based on variable length *k*-mers, preferring frequently occurring longer *k*-mers [20], and displays high recall but at the cost of numerous false positives. GI-SVM [21] and Wn-SVM [22] both use one-class SVM to identify GIs, but the difference is that the GI-SVM utilizes windowed genome sequence, whereas the Wn-SVM works at the gene level. GIHunter [23], on the other hand, is a supervised learning algorithm, aiming to overcome the limitations of other similar approaches by exploiting the functional annotations of the genes. Since it utilizes functional and structural information, contrary to many other algorithms that often work with the raw sequence alone, GIHunter is comparatively more accurate in GI prediction. However, the promise of this class of methods hinges on robust functional annotation, which may not be forthcoming with newly sequenced genomes, particularly those belonging to taxa that are less understood. Moreover, these functional annotation-based methods may miss GIs that have lost the functional marker genes.

In addition to developing GI prediction algorithms, significant efforts have been made in building databases of GIs using combinations of existing methods. The Islandviewer [24, 25] project employs an ensemble of two sequence-based methods, SIGI-HMM [15] and IslandPath-DIMOB [17], an annotation-based method Islander [26] and a comparative genomics method IslandPick [14] to compile a list of GIs in each prokaryotic genome. Predictions for most of the well-known genomes are precomputed and the database is updated regularly [25]. IslandPath-DIMOB [17] provides the researchers with an interactive graphical user interface to aid in the visualization of predicted GIs. The mobility genes, tRNA genes and genes with atypical GC content and dinucleotide bias are identified, followed by supervised manual curation. Some other notable GI databases are MOSAIC [27], Islander [28], PAIDB [29], etc.

In this paper, we present a novel method for identifying GIs in prokaryotic genomes, which uses only the unannotated sequence from a single genome. We propose here an unsupervised machine learning-based method, SSG-LUGIA, which obviates the need of functional annotation of the genome for localizing GIs. Unsupervised learning approaches have

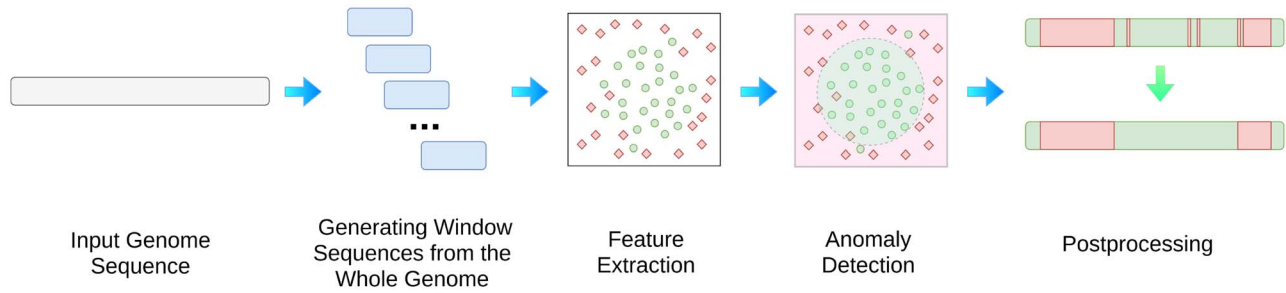


Figure 1. Schematic of the algorithmic pipeline of SSG-LUGIA. SSG-LUGIA takes a genome sequence as input, splits it into overlapping windows, computes features from the windowed sequences and identifies the anomalous (potentially horizontally acquired) windowed sequences thereby. Finally, the output is further refined following a postprocessing routine.

advantages over supervised methods particularly when the data are heterogeneous, which are often so with versatile prokaryotic genomes, and therefore, the models trained on GIs from a set of genomes in a supervised setting may not be generalizable to identify novel GIs in yet unknown genomes [30]. Also, SSG-LUGIA's nondependence on functional annotation, as opposed to several well-established methods [23], makes it useful particularly in analyzing just sequenced, anonymous genomes. SSG-LUGIA is based on an anomaly detection procedure that relies on the atypical compositional biases of the alien genes. Moreover, the results are postprocessed based on concepts from signal processing with a goal to significantly reduce noisy predictions. SSG-LUGIA outperformed several frequently used existing methods on the IslandPick benchmark dataset [14]. SSG-LUGIA was also assessed on the revised IslandPick dataset [31] and was compared with 20 other methods. Furthermore, SSG-LUGIA achieved superior results compared with other considered methods on the well-characterized *Salmonella typhi* CT18 genome. The efficacy of SSG-LUGIA in identifying laterally transferred genes was also evaluated on two additional well-understood genomes of gram-positive and gram-negative bacteria, namely, *Corynebacterium diphtheria* NCTC13129 and *Pseudomonas aeruginosa* LESB58. Notably, SSG-LUGIA balances the precision and recall well, in contrast to most of the existing methods. Moreover, by simply tuning the parameters, SSG-LUGIA provides the flexibility of prioritizing precision, recall or ensuring an acceptable balance between these two accuracy metrics based on the need of a particular application. SSG-LUGIA thus advances the state of the art in GI and alien gene detection. SSG-LUGIA has been released as an open-source software to aid researchers in analysis and visualization of foreign elements in prokaryotic genomes.

Methods

We first present a brief overview of SSG-LUGIA that takes an unannotated genome sequence as input and predicts GIs and optionally enlists putative horizontally transferred genes in case a gene annotation file is provided. The genome sequence of interest is first split into overlapping window frames that are then analyzed for the presence of GI distinguishing features compiled from publicly available literature on GIs. The genome sequence is screened for 'anomalous' regions based on GI distinguishing features using an unsupervised machine learning procedure. The anomalous segments thus identified are further refined following a postprocessing step, and finally, the proximal segments are merged to produce the list of GIs (and optionally, the list of horizontally transferred genes). The algorithmic pipeline is illustrated in Figure 1.

Generating window sequences from the whole genome

At first, the whole-genome sequence is split into overlapping subsequences by sliding a window over the genome sequence, similar to the windowing approach of GI prediction tools such as, GI-SVM [21], Centroid [32] and Alien Hunter [20]. Splitting the whole sequence by sliding a window of length $window_{len}$ with a step size of $window_{step}$ enables analysis of genomic signatures in each individual segment and thus aid in distinguishing between the native and alien regions. However, it should be noted that the performance of a moving window method is sensitive to the window size, and therefore, $window_{len}$ needs to be chosen carefully. Too small window size diminishes signal-to-noise ratio due to a smaller context, whereas a large window size reduces resolution, missing not just smaller GIs but also effectively flattening the irregularities critical in identifying the alien regions.

Feature extraction

GC composition

The fact that the GIs originate from sources with different genomic contexts than that of the host organism, compositions of acquired sequences differ from that of the host genome. The most straightforward and widely used measure of compositional bias is the GC content. GC Content is obtained as proportion of G and C in a genome sequence of interest (Equation 1). GC compositional bias has been exploited in a number of GI or alien gene prediction algorithms [16, 33–36].

$$GC \text{ Content} = \frac{Count(G) + Count(C)}{Count(A, T, G, C)} \quad (1)$$

Similar to GC Content, GC Skew can also be a promising trace of alien regions [37, 38], defined below.

$$GC \text{ Skew} = \frac{Count(G) - Count(C)}{Count(G) + Count(C)} \quad (2)$$

Therefore, we consider GC Content and GC Skew among compositional features to discriminate potentially alien regions from native regions.

K-mer features

Most of the genome sequences have a stable distribution of k -mers to the extent that k -mers may be termed as the barcode of a genome [39]. Therefore, frequency distributions of k -mers in different regions of the genomes has previously been exploited to detect alien regions in a genome [21, 32, 40].

Dinucleotide bias (2-mer):

We use the dinucleotide bias metric, primarily proposed by Karlin [41]. However, we slightly change the formulation by taking the square-root value of the product of the denominator terms, as follows.

$$2\text{-mers} = \left\{ \frac{\text{freq}(XY)}{\sqrt{\text{freq}(X) \times \text{freq}(Y)}} \mid X \in [A, T, C, G] \wedge Y \in [A, T, C, G] \right\}, \quad (3)$$

where $\text{freq}(XY)$ and $\text{freq}(X)$ denote the frequencies of dinucleotide XY and nucleotide X , respectively.

Codon bias (3-mer):

Instead of the 3-mer compositional bias, we considered codon usage bias for each amino acid as follows [42]:

$$3\text{-mers} = \left\{ \sum_{k \in \mathbb{K}_3} \text{freq}(k) \mid \text{Translate}(k) = a \wedge a \in \mathbb{A} \right\}, \quad (4)$$

where $\mathbb{A}$ and $\mathbb{K}_3$ are sets of all amino acids and 3-mers, respectively.

Higher-order k-mers:

Tetranucleotide (4-mer) frequencies have previously been shown to be an effective determinant of atypicality in genomes [38]. Therefore, we also computed the frequencies of the 4-mers and stored them in a vector.

$$4\text{-mers} = \{\text{freq}(k) \mid k \in \mathbb{K}_4\}, \quad (5)$$

where $\mathbb{K}_4$ is the set of all 4-mers.

Although various methods have considered k -mers of $k = 2 \dots 9$ [8, 21], in our experiments we observed that including the higher-order k -mers ($k > 4$) does not improve the performance much compared with the additional computation they demand. Therefore, we used k -mer features for $k = 4$ in SSG-LUGIA, and the lower-order features were accounted for through dinucleotide and codon usage biases as described above. Notably, unlike various methods like SIGI-HMM, which compute k -mer frequencies or codon usage bias at the gene level, SSG-LUGIA computes the k -mer frequencies from windowed genomic sequences. Thus, SSG-LUGIA obviates the need of gene annotation to localize GIs.

Dimensionality reduction

A rather unfortunate consequence of using k -mer frequencies as features is that the data become high dimensional. In addition to being high dimensional (e.g. for $k = 4$, we have $4^4 = 256$ features), the respective feature vectors are likely to be sparse, as not all k -mers occur frequently in the sequence. This so-called 'Curse of Dimensionality' is detrimental for machine learning algorithms [43]. Therefore, to make our learning process easier and predictions more reliable we perform dimensionality reductions on the k -mer features.

Principal component analysis (PCA) forms the basis for multivariate data analysis involving a large number of dimensions [44]. PCA, in its simplest, identifies new variables from the linear combinations of existing variables in the data, using Eigenvectors and Eigenvalues, obeying the rules of linear algebra [45]. These new variables, denoted as principal components, are sorted in terms of significance and only a few of the most significant variables are usually sufficient to describe the patterns and trends of the entire, otherwise high-dimensional dataset [46].

We performed PCA on dinucleotide bias, codon bias and 4-mer frequency-based features individually and selected the most significant components. Another desirable outcome of applying PCA over these sets of features is that PCA normalizes the feature values decently, which facilitates training and prediction.

Entropy features

Entropy-based features have also shown great potential in the detection of laterally transferred genes [30, 47]. Most notably, JS-N, JS-DN and JS-CB have attained significant efficacy in distinguishing alien sequences from the host genome [30]. JS divergence [48] between two probability distributions P_1 and P_2 of a discrete random variable is defined as,

$$JS_\pi(P_1, P_2) = H(\pi_1 P_1 + \pi_2 P_2) - \pi_1 H(P_1) - \pi_2 H(P_2). \quad (6)$$

Here, π_1 and π_2 are weight factors, obeying the relation $\pi_1 + \pi_2 = 1$. $H(\cdot)$ denotes Shannon information entropy, defined as

$$H(P) = - \sum_i P(i) \log_2 P(i), \quad (7)$$

where, $P(i)$ is the probability of the i th element of the distribution P .

However, JS divergence features convey a pairwise divergence measure between two distributions. In our case, we consider the distribution of the features in each individual region and the mean of those across the whole genome. But since we are applying an anomaly detection pipeline on the computed features, the contrast with the mean is redundant in the feature computation stage. Therefore, we only compute the entropy values $H(\cdot)$ for the frequency distributions of 2-mer, 3-mer and 4-mers and use those values as our entropy features. It should be noted that the entropy values are computed before performing the PCA-based dimensionality reductions on the k -mer frequencies.

Anomaly detection

The primary rationale behind using compositional and entropy features to distinguish alien sequences from native sequences of the host genome is that they are likely to be atypical relative to the native sequences in these features. From machine learning perspectives, alien sequences can be identified by a supervised classifier trained on sets of features extracted from verified alien and native sequences. However, as pointed out in the previous sections, this approach may not be appropriate for two reasons: (i) there are not enough validated alien and native sequences to train a classifier upon and (ii) the evolutionary consequences may impart significant heterogeneity making detection of novel elements particularly challenging for these methods. We, therefore, have developed an unsupervised approach that relies on comparing the characteristics of the genomic regions with the mean characteristics of the overall genome (i.e. average over all genomic regions), similar to several notable studies in the field [41, 49, 50]. The novelty of our approach is reflected in an appropriate unsupervised machine learning pipeline for anomaly detection.

In terms of statistical machine learning, comparing a sample data point with the mean of the data distribution reduces to calculating the Mahalanobis distance [51] of that point from the centroid of the data. Mahalanobis distance has been being

popularly used in outlier/anomaly detection in multivariate data in many different domains [52, 53]. The idea here is straightforward: the higher the distance, the more divergent the data point is from the distribution. Therefore, by comparing with a predefined threshold or by considering a percentage of the farthest data points, one can identify probable outliers in the data distribution.

Mathematically, let $\vec{x} = (x_1, x_2, x_3, \dots, x_d)$ be a data point in a d -dimensional data distribution X . The centroid of the distribution $\vec{\mu}$ is the arithmetic mean of the data points along all the diagonals. The Mahalanobis distance of data point $\vec{x}$ is computed as:

$$D_M(\vec{x}) = \sqrt{(\vec{x} - \vec{\mu})^T S^{-1} (\vec{x} - \vec{\mu})}, \quad (8)$$

where, S is the covariance matrix of the data distribution X .

After computing the Mahalanobis distance for all the data points, a contamination threshold between 0 and 100 is selected and the points with Mahalanobis distances in the top contamination percentile are identified as outliers. After extracting the features for windowed sequences as described above, we perform anomaly detection based on Mahalanobis distance. However, it was observed that even after identifying the outliers, a substantial number of alien regions were still identified as native regions. Interestingly, performing the anomaly detection procedure again on the already filtered windowed sequences resulted in significant reduction in false negatives, thus improving the recall significantly. To further improve the sensitivity, we adopt a two-step anomaly detection using two thresholds $contamination_1$ and $contamination_2$.

Postprocessing

Detecting anomalies through a two-stage process substantially increases recall but at the cost of a large number of false positives. To augment precision, we perform the following postprocessing. Instead of taking binary outputs from the anomaly detection algorithm, we extract the values of the Mahalanobis distances and subtract the offset, i.e. the centroid location. Analyzing the distance values (after the above-mentioned subtraction) reveals some sudden, unexpected spikes in the native regions. Such sudden spikes are known as impulse function in signal processing literature and median filters are used to remove them [54]. A window length for the filter, $filter - len$ is selected and the filtered distance is obtained as follows.

$$\begin{aligned} \text{Filtered Distance}[i] &= \text{median}(\text{Raw Distance} \\ &\quad [i - \frac{filter - len}{2} : i + \frac{filter - len}{2}]), \quad (9) \\ i &= 1, 2, \dots, len(\text{Raw Distance}) \end{aligned}$$

A threshold is then applied to the filtered distance values and the anomalies are thus refined through smoothing the noisy predictions. It should be noted that to keep the computations coherent with standard anomaly detection pipelines of the Scikit-learn library [55], we considered the negative values of the distances to predict potentially alien regions, i.e. the distance of the distant anomaly points be less than zero. Thus, a threshold

is set as follows:

$$\begin{aligned} \text{AnomalyDetection}(x) &= \\ \begin{cases} \text{Alien Region} & \text{if Filtered Distance}(x) < 0, \\ \text{Native Region} & \text{if Filtered Distance}(x) \geq 0. \end{cases} \quad (10) \end{aligned}$$

Merging the window sequences and horizontal gene identification

Finally, the predictions of the windowed sequences (contiguous atypical windows) are merged together and the GIs along the whole genome are determined. Although smoothing the noisy predictions significantly reduces false positives and improves precision, a handful number of scattered small predicted islands are still inferred erroneously. Owing to the fact that the alien genes clustered within GIs tend to cover quite a fairly long region [56], we discard regions smaller than a certain threshold $min_{island-length}$. Elimination of such falsely predicted smaller regions considerably improves the precision of our method. Finally, SSG-LUGIA returns the coordinates and lengths of probable GIs in a query genome sequence. Optionally, if gene annotation is provided along with the genome sequence, SSG-LUGIA also returns the list of likely horizontally transferred genes.

Parameter tuning

SSG-LUGIA uses a number of tunable parameters, proper tuning of which is crucial for application thereof. To tune the parameters appropriately, a reliable and rich dataset of GIs is required, which is lacking due to the difficulty or infeasibility of validating evolutionary events [57]. Therefore, we used the IslandPick [14] benchmark dataset to tune the parameters of SSG-LUGIA (more details on IslandPick are provided in the following section). However, since we evaluated our algorithm on this benchmark dataset, this tuning was done cautiously to avoid overfitting our model to this dataset. Hence, we designed our parameter tuning task as follows: 20 genomes were sampled randomly each time and the performance was tested on the sampled genomes for a certain set of parameters. These sets of parameters were selected through an exhaustive grid search; for example, for the PCA features we tried different combinations of features. Since we continuously shuffled and used only 20 genomes from 118 genomes in each round, it is quite reasonable to assume the process being largely devoid of any biases and overfitting.

Receiver operating characteristic (ROC) curve is a popular choice for threshold selection or parameter tuning. However, there exists a drawback of ROC curves when the class distribution is skewed as we have for GIs. In the presence of class imbalance, precision-recall (PR) curve is more suitable [58]. Therefore, we compute the precision and recall scores for the individual sets of parameters and plot a PR curve as shown in Figure 2.

The points on the pareto-front are the most interesting and important ones for parameter selection. In our experiments, we found several points on the pareto-front with reasonable precision. Among them, we selected the corner point (colored in Cyan) as our preferred model (Figure 2).

We focused on the precision to make our method more reliable. However, for certain applications, higher recall is more important so that no vital GIs are lost. This has led to high recall methods like GI-SVM [21] and AlienHunter [20] with relatively worse precision. In order to offer a similar high recall version to satisfy application-specific requirements, we selected the

Table 1. Parameters used in the different variants of SSG-LUGIA. In order to establish the optimal parameter settings, extensive experimentation was performed by randomly picking and tweaking the values of the different parameters. Instead of devising a single model, we ultimately developed three models, emphasising on recall, precision and F1-score, respectively, suiting the specific application needs

Model	window _{len}	window _{step}	pca _{2-mer} pca _{3-mer} pca _{4-mer}	contamination ₁	contamination ₂	filter-len	min _{island-length}
SSG-LUGIA _R	10000	100	2	0.2	0.25	400	10000
SSG-LUGIA _P				0.075	0.075		
SSG-LUGIA _F				0.15	0.05		

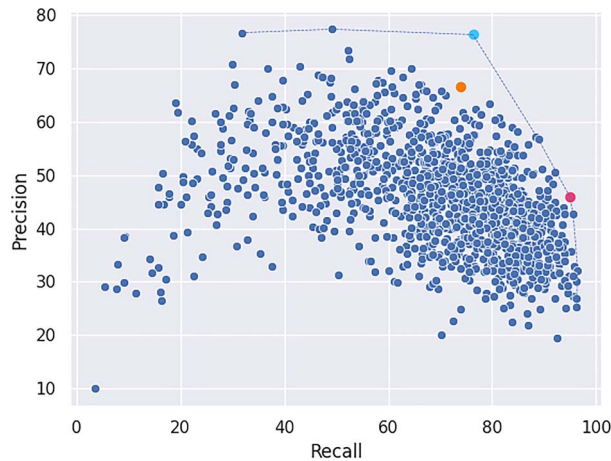


Figure 2. Parameter selection. We performed a grid search on the parameters and randomly selected a set of 20 genomes for each parameter set. The precision and recall values for each configuration have been computed and a corresponding PR curve has been plotted. From the configuration points, we select three interesting parameter sets: SSG – LUGIA_P, focused on precision (colored in cyan), SSG – LUGIA_R, focused on recall (colored in red) and SSG – LUGIA_F, achieving the best F1-score during tuning (colored in orange).

knee point' (colored in red in Figure 2) from the pareto-front as after that point, although the recall slightly increases, the precision sharply declines. We call these two variants SSG – LUGIA_P and SSG – LUGIA_R, based on their focus on precision and recall, respectively. The values of the parameters are provided in Table 1.

In our random parameter search, we observed that some sets of parameters contribute equivalently to high precision and high recall scores (e.g. window_{len}, window_{step}, pca_{2-mer}, pca_{3-mer}, pca_{4-mer}). The rest, on the other hand, must be considered in a precision-recall trade-off. contamination₁ and contamination₂ are directly correlated to recall, as increasing them almost always increased the recall but at the cost of precision. Higher values of filter – len_{median} improved precision but reduced the recall. Increasing min_{island-len} consistently improved precision, which could be due to the fact that the GI datasets could be biased toward longer islands; furthermore, shorter GIs are prone to be misclassified. Therefore, in order to keep the algorithm flexible, min_{island-len} was set to 10 000 bp following [56]. It should be noted that this is also consistent with the selected window length of 10 000 bp.

In addition to precision and recall focused variants of our method, we also observed that a particular parameter setting yielded the best F1-score in the tuning process (colored in orange). However, it excels in neither precision or recall scores like the other two variants. We refer to this version as SSG – LUGIA_F, and upon further evaluation we observed that

although it manages to balance well the precision and recall, may not always yield optimal overall performance.

Experiments and results

Exploring feature significance

Efficacy of different features in distinguishing alien regions from native ones has previously been assessed. However, the results have not been consistent, rather they demonstrate a high degree of variance [30]. This motivated us in evaluating the contributions of various features used in SSG-LUGIA to its performance in detecting horizontally acquired regions. In an unsupervised setting, it is nontrivial to infer the feature significance, particularly when many features are involved. Supervised methods such as, Random Forest [59], which is an ensemble of decision trees, can readily compute importance scores of the features [60] used in classification (Section 2 in Supplementary Material SM1). We, therefore, trained a random forest classifier, comprising 10 decision trees, on individual genomes in the IslandPick dataset. Contributions of different features in resolving the distinction between native and alien regions for the individual genomes were computed and the overall trend was observed. The relative importance obtained in this way is shown in Figure 3A. GC skew, codon usage bias and the PCA components obtained from dinucleotides are among the most prominent features. Nevertheless, it is also apparent that all features contribute comparably when tested on a variety of genomes. Thus, this study provides an empirical verification of the uncertainty of feature selection in determining GIs as reported in previous studies [30]. Therefore, instead of focusing on a subset of these features, we aimed for analyzing them in congression to achieve reliable predictions.

Furthermore, to assess the contributions of the different features, we experimented with a leave-one-out scheme. We started with the baseline SSG – LUGIA_F model and repeatedly removed a certain type of feature and computed the performance. The change of metric scores can thus be attributed to the contribution of that particular feature. The computed contributions of the individual features are presented in Figure 3B. From this figure, we can observe that removing any of the features reduces the recall and slightly improves the precision. This is expected, as in [30] it was demonstrated that different sets of features aid detection of GIs in different genomes. Thus, when we remove a feature, its impact could reflect either its redundancy or necessity. On the genomes or specific GIs for which it is a necessary feature, its absence will result in misidentification of the corresponding GIs, i.e. lower recall. On the contrary, for those genomes where it is redundant, ignoring that feature will likely not alter the outcomes. Removing such features and those that may even confound classification when used in combination with others may simplify the model making it a more robust predictor on yet unseen data. Precision

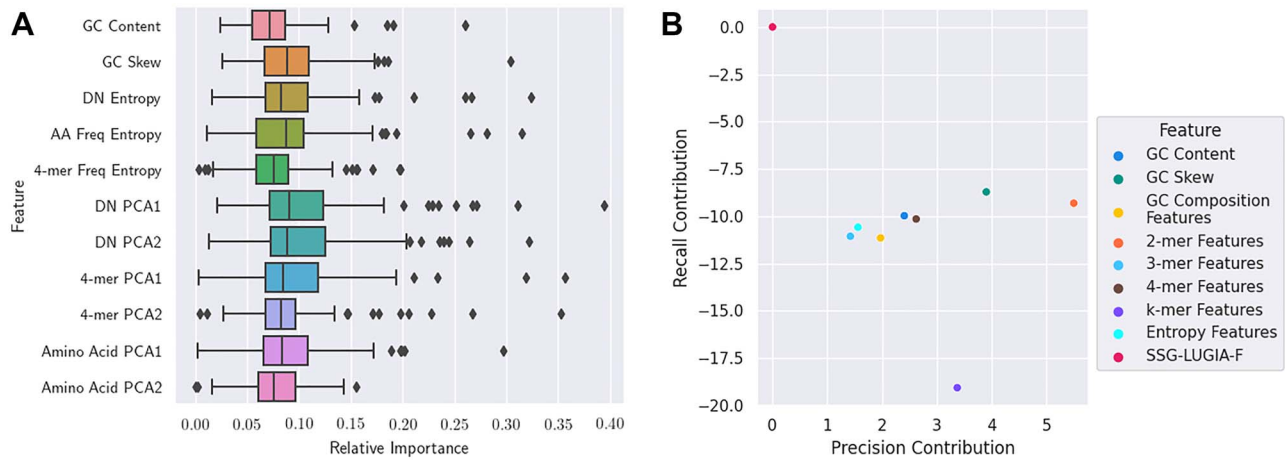


Figure 3. Exploring feature significance. (A) presents the relative prominence of various features in discriminating the alien regions from native ones. The features were used to classify the alien and native regions via supervised learning of Random Forests, and the feature importance was computed on all the genomes from the IslandPick dataset. (B), on the other hand, presents the individual contributions in precision and recall of different features. In this experiment, we considered SSG – LUGIA_F as the baseline model and ran it by removing different features, following a leave-one-out scheme. We show the change (%) in precision and recall in x- and y-axis, respectively. Removing a feature increases precision (i.e. removes false positives corresponding to that feature) and decreases recall (i.e. removes corresponding true positives). Thus, the contributions of individual features can be inferred from B.

may further improve as these features are likely to contribute to false positive predictions. The recall-precision contribution plot highlights the contributions of the features ranked by Random Forest to the recall and precision (Figure 3A). GC Skew and PCA components obtained from dinucleotides contributed the most to the precision, which were also ranked high by importance in the discrimination in the Random Forest-based experiments.

Evaluation of GI prediction on IslandPick benchmark dataset

Evaluating algorithms for biological data analysis is often challenging due to the lack of benchmark datasets. Previously, methods for GI prediction relied either on artificial genomes [7] or on a handful of well-understood genomes [21]. However, Langille et al. [14] performed a study with a broad set of 118 genomes and six algorithms. Using a comparative genomics approach, a benchmark was hence formulated, which was used for the comparison of various methods.

In this benchmarking, true positives (TP) refer to known GI nucleotides that are predicted correctly by a method. False positives (FP) refer to known non-GI nucleotides that are incorrectly predicted as part of GIs. In the similar vein, true negatives (TN) refers to known non-GI nucleotides that are correctly predicted by a method and false negatives (FN) refer to GI nucleotides that are incorrectly predicted as non-GI nucleotides. Then, the recall, precision, F1-score and accuracy values were calculated using the standard formulae (Section 1, Supplementary Material SM1). However, in the original benchmarking work [14], the authors presented the average values of these metrics using the average values of TP, TN, FP and FN from the entire dataset. This type of computation may lead to false perception over performance on individual genomes, as the results would be biased by differential size of genomes and also the number and size of GIs in each genome. Therefore, we calculated the average values of the metrics as the average of the values obtained for the individual genomes (the values of metrics for the individual genomes for other methods were collected from Additional file 6 of [14]). The results for the various algorithms along with the three

Table 2. GI prediction performance on the IslandPick benchmark dataset. We present a comparison with the existing methods, which were evaluated in this benchmark. Instead of averaging the metrics over the entire dataset, we average them over individual genomes to avoid inflating the values by longer genomes and/or GIs. Best results for each evaluation metric are shown in bold, and the second best results are italicized

Method	Precision (%)	Recall (%)	F1 (%)
SIGI-HMM [15]	75.783	32.849	42.0314
Centroid [32]	43.655	23.587	27.0367
PAI IDA [16]	53.945	33.054	35.907
IslandPath/DIMOB [17]	64.284	36.22	41.518
IslandPath/DINUC [17]	45.818	57.324	45.740
Alien Hunter [20]	38.619	81.091	46.184
SSG-LUGIA _R	40.011	86.755	49.062
SSG-LUGIA _P	60.280	55.044	52.376
SSG-LUGIA _F	55.105	65.223	55.079

different variants of the proposed SSG-LUGIA are summarized in Table 2.

Of the six methods studied in this benchmarking, SIGI-HMM [15] scores the highest precision (~ 75%, Table 2). However, this comes at a cost of (unacceptably) low recall (~ 33%). Island-Path/DIMOB [17–19] comes in 2nd in terms of precision (~ 64%), yet with a poor recall (~ 36%). The precision-oriented variant of our proposed SSG-LUGIA_P, despite achieving the 3rd best precision score (~ 60%), scores 18 – 22% higher recall than the more precise methods. The F1-score oriented SSG-LUGIA_F also scores a moderate precision. On the other hand, SSG-LUGIA_R achieves the highest recall (~ 87%) though by a significant reduction in precision (~ 40%). Nevertheless, the precision of even SSG-LUGIA_R is almost close to that of Centroid and IslandPath/DINUC, which is remarkable considering its high recall (Table 2).

On the other hand, all current methods except Alien Hunter [20] perform poorly in terms of recall. Apart from Alien Hunter (~ 81%) and IslandPath/DINUC (57%), the recalls of the remaining current methods are below 40%, indicating that these methods sacrifice their recall in order to achieve high precision. However,

even the precise version of SSG-LUGIA_P scores a recall of ~ 55%, which is superior to the recalls of most other methods. SSG-LUGIA_F scores even higher recall, and the recall-oriented SSG-LUGIA_R scores the highest recall (~ 87%) (Table 2).

One of the most notable observations (Table 2) is the trade-off between precision and recall, which is apparent from the results of [14] as well. The methods with higher precision have a low recall, and the methods with higher recall have a low precision. However, as it is apparent from Table 2, SSG-LUGIA is comparatively less skewed in this aspect, despite being a bit biased toward attaining higher recall. This fact is also evident from the high F1-scores of SSG-LUGIA. Most notably, SSG-LUGIA_F achieves the highest F1-score of ~ 55%. Moreover, a comparison between SSG-LUGIA_R and Alien Hunter shows that the former outperforms the latter in every metric as reported in Table 2 (Figure S4 in Supplementary Material SM1, where we present box plots, showing values of various performance metrics for these methods on the IslandPick benchmark dataset).

We have also provided the accuracies for each of the 118 individual genomes in IslandPick dataset (Supplementary Material SM2). Out of these 118 genomes, SSG-LUGIA variants outperformed other methods on 88, 31 and 59 genomes in terms of recall, precision and F1-score, respectively, and achieved the 2nd best accuracy on 67, 40 and 54 genomes in terms of recall, precision and F1-score, respectively.

Comparison with a broader set of methods

Owing to the profound impact of GI on bacterial evolution and therefore the need to robustly delineate GIs in bacterial genomes, over the years a number of methods have been introduced. Some methods like AlienHunter [20] and MTGipick [61] put emphasis on recall, whereas methods like Zisland Explorer [62] and Islander [26] prioritize precision over recall. These outcomes reflect the desire to be either conservative/stringent (i.e. high precision) or more sensitive (i.e. high recall). These methods can be further classified based on their pipelines and features. Methods that leverage the functional and structural features tend to be more reliable as enrichment of features such as mobility or tRNA genes provides compelling evidence in support of the predictions [31]. On the contrary, methods relying on only sequence-based features tend to be more inclusive but less precise. Hybrid methods such as IslandViewer 4 [25] that combines predictions of IslandPath-DIMOB [19], SIGI-HMM [15], Islander [26] and IslandPick [14] exploit diverse features and evidence to comprehensively catalog GIs [31].

The IslandPick dataset has been recently revised by Bertelli and Brinkman [19] as an updated benchmark for the GI prediction tools. They considered 37 recent methods and assessed the performance of 20 methods on 104 genomes. Therefore, to compare SSG-LUGIA with a broader set of methods, we evaluated the performance of SSG-LUGIA on those genomes as well. However, a fair comparison is not possible as the methods are vastly different even merely based on the input data. For example, methods that utilize structural features may delineate GIs well unless such features have degraded over time [31]. On the other hand, methods that solely depend on sequence composition information tend to be more sensitive but generate numerous false positives. Hybrid methods are designed to exploit the complementary strengths of multiple methods, while bypassing their pitfalls. For a fair comparison, we grouped these methods into these three categories (Figure 4). The values of the performance metrics for current methods assessed on this dataset were obtained from Bertelli et al. [31].

Table 3. Performance evaluation of GI prediction methods on the *S. typhi* CT18 genome

Method	Recall (%)	Precision (%)	F1 (%)
GI-SVM	89.5	44.6	59.6
EGID	77.9	53.5	63.4
SIGI-HMM	24.1	55.6	33.7
IslandViewer	65.4	67.0	66.2
GIHunter	82.7	67.6	74.4
IslandPath-DIMOB	55.3	78.8	65.0
tRNAcc	28.6	99.3	44.4
IslandPick	6.0	100.0	11.4
SSG-LUGIA _R	92.5	39.7	55.6
SSG-LUGIA _P	66.3	89.7	76.2
SSG-LUGIA _F	77.6	75.4	76.5

From these results, it is evident that despite utilizing only sequence composition information, SSG-LUGIA performs remarkably well. Notably, SSG-LUGIA_R outperforms hybrid and structural feature-based methods in terms of recall, obtaining the highest recall of 75.3%. SSG-LUGIA_R also achieves the 3rd best F1-score, after hybrid IslandViewer 4 and structural feature-based supervised method GIHunter. Although SSG-LUGIA_F does not outperform all of hybrid and functional feature-based methods, it achieves a better precision-recall balance than other sequence composition-based methods. Assessment on another broad benchmark dataset reiterates that SSG-LUGIA is a useful sequence composition-based GI prediction tool, especially for newly sequenced, yet uncharacterized genomes.

Case study: *S. enterica* serovar *typhi* (*S. typhi*) CT18 genome

Salmonella typhi CT18 is a gram-negative pathogen that is responsible for gastroenteritis and typhoid fever [63]. As the CT18 strain has been well studied and characterized particularly in the context of presence of GIs [57, 64], we evaluated the performance of SSG-LUGIA on the CT18 genome as well. The GI annotation and performance of other algorithms were collected from [57].

Comparative performance assessment is presented in Table 3. Although the precision for IslandPick, tRNAcc and IslandPath-DIMOB is high, their recall is significantly low. SSG-LUGIA_P on the other hand, achieves a moderately high precision with a relatively better recall. In terms of the overall accuracy (F1-score), SSG-LUGIA_F is the best performing method. SSG-LUGIA_P also compares well in this aspect, as it produces the second best F1-score. It should be noted that unlike GIHunter and some other programs, SSG-LUGIA attains this level of performance independent of marker gene information, a remarkable feat considering that it used only compositional information. Clearly, SSG-LUGIA_F balances recall and precision well, achieving an overall accuracy of ~ 77%. On the other hand, the highest recall score was obtained by SSG-LUGIA_R (~ 93%).

Figure 5 shows the GIs predicted by various methods as well as the 19 known GIs in the CT18 genome (omitting the two islands smaller than 5000 bp, as previously done in [57]). This example illustrates how some GI prediction methods, such as GI-SVM, make many predictions that include false positives, whereas others, such as IslandPath-DIMOB, SSG-LUGIA_P and GIHunter, make limited but precise predictions. From the GIs predicted by various methods in Figure 5, it is evident that islands predicted by SSG-LUGIA_F are most identical to the ground truth with relatively minimal false positives. Moreover,

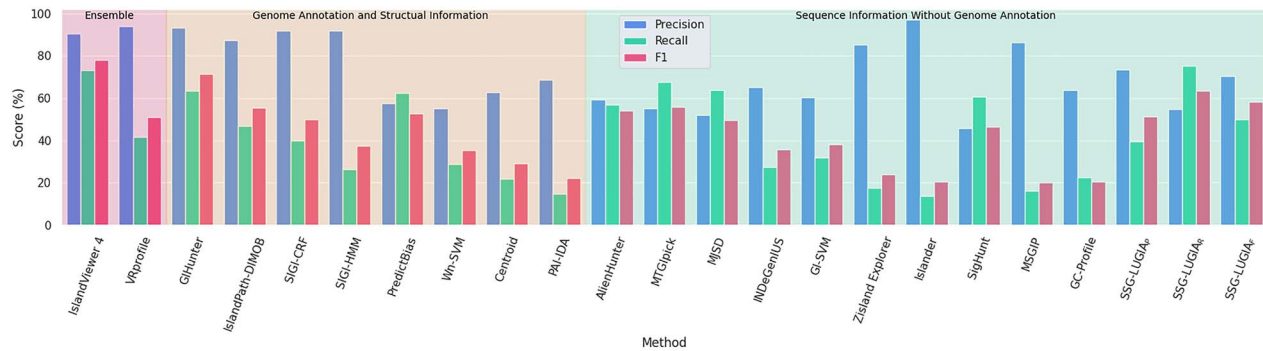


Figure 4. Comparison of SSG-LUGIA with a broader set of methods. We group different methods into three categories (ensemble methods, methods, which use genome annotation and structural information, and sequence-based methods). SSG-LUGIA does not only excel over other methods in its category, but it is competitive with methods from other categories as well.

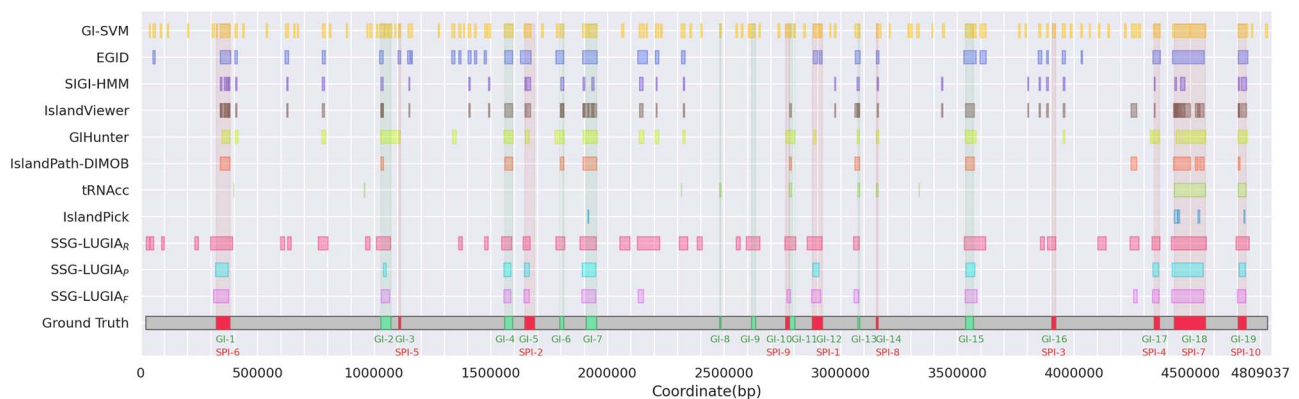


Figure 5. GIs predicted by various methods in the CT18 genome. In total, 19 known GIs are shown on the bottom as 'ground truth'. In addition, we put focus on the well-studied PAIs (10 islands out of these 19). Here, the PAIs are shown in red, and the remaining GIs are shown in green.

other than a few small islands, most of the islands are predicted by SSG-LUGIA_R, and remarkably, SSG-LUGIA_P has not predicted any false GIs.

PAI is special type of GI that harbors genes encoding virulence factors. Among the 19 GIs in the CT18 genome, 10 are PAIs, which could be of interest, particularly in the context of virulence trait of this pathogen. The PAI information is collected from PAIDB [65] and their locations are marked in red color in Figure 5. SSG-LUGIA_R identified 8 of the 10 PAIs. GI-SVM is the only method that could identify all the PAIs; however, it generated many false predictions and thus could not attain a good balance in the precision-recall trade-off (Figure 5). We also examined the overlap between PAIs and respective predictions from different methods (Figure 6). It is evident that although SSG-LUGIA_R misses two small PAIs, it covers the detected PAIs with much higher overlaps in comparison to GI-SVM or GHIHunter. The balanced version SSG-LUGIA_F also performed comparably in this aspect (Figure 6).

Additional results

Evaluation of methods for laterally transferred gene identification

In addition to assessing the effectiveness of identifying GIs at the nucleotide level, we also appraised the efficacy at the gene level. If gene annotation is provided along with genome sequence, SSG-LUGIA also outputs a list of probable laterally transferred genes harbored by predicted islands. The efficacy of SSG-LUGIA

in identifying horizontally transferred genes was evaluated on two additional bacterial genomes, namely, those of *C. diphtheria* NCTC13129 and *P. aeruginosa* LESB58. SSG-LUGIA achieved the best F1-score on NCTC13129 genome (10.9% higher than the 2nd best method). On LESB58, the F1-score achieved by SSG-LUGIA_P (65.6%) was 2nd only to Wn-SVM (68.7%). SSG-LUGIA_P achieved the highest precision on both NCTC13129 and LESB58, and SSG-LUGIA_R achieved the highest recall on NCTC13129 (see supplementary material SM1 for more details).

GI Prediction from Draft Genomes

Since SSG-LUGIA solely works on sequence without any functional annotation, it is suitable for predicting GIs in draft genomes. We performed a draft genome simulation study similar to GI-POP [66]; SSG-LUGIA_F consistently outperformed GI-POP for any percentages of the nucleotides removed from the test genomes (Supplementary Material SM1).

SSG-LUGIA as a component of an ensemble method

Ensemble methods like IslandViewer 4 [25] combine the strengths of multiple methods and use a wide array of features (e.g. functional annotation, sequence composition, etc.) and may result in a better performance compared to that of standalone methods like SSG-LUGIA. To further assess the efficacy of SSG-LUGIA, we explored the possibility of SSG-LUGIA being used in a

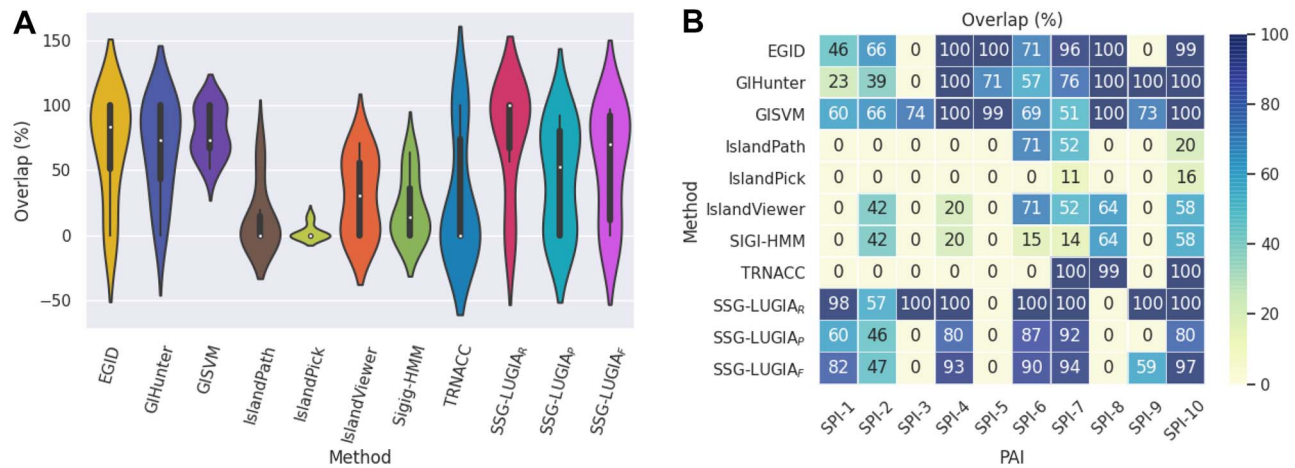


Figure 6. Degree of overlap between the PAIs and the predictions made by different methods. **(A)** Violin plot showing distribution of PAI coverage by several methods. **(B)** Heatmap showing percentages of the known PAIs contained within the islands predicted by different methods.

hybrid method like IslandViewer 4. From our analysis on 15 well-studied genomes, we found SSG-LUGIA to be suitable for use in an ensemble method and, notably, an ensemble of SSG-LUGIA_r, Islander and IslandPath-DIMOB outperformed IslandViewer 4 that is an ensemble of four methods (Section 8 in Supplementary Material SM1).

Evaluation on 15 well-studied genomes

We further assessed the performance of SSG-LUGIA on 15 well-studied genomes. Ten genomes among these were not included in other datasets analyzed in this study, and thus they provided us with further scope for independent testing. Results on these 15 genomes are presented in Supplementary Materials. SSG-LUGIA_f compared well with 11 other methods tested in this benchmark, outperforming 9 of the 11 methods. IslandCafe [67] and IslandViewer that outperformed SSG-LUGIA_f utilize functional annotation and ensemble approach, respectively, unlike SSG-LUGIA that is based solely on sequence composition.

Additional remarks on the utility of sequence-based methods and the impact of functional annotation on GI prediction

We systematically analyzed the impact of functional annotation on the performance of a recent, state-of-the-art method IslandCafe, which is an improvement over GEMINI [68] and combines multiple prevalent GI inference concepts including phylogenetic- and annotation-based information [67] (Section 9 in Supplementary Material SM1). Our study provides evidence that a major challenge impacting the utility of current annotation-based methods is their dependency on the availability of reliable genome annotations, making their application difficult on newly sequenced, uncharacterized genomes. In contrast, methods such as SSG-LUGIA use sequence composition only. Our results show that GI prediction would benefit from renewed attention toward improving sequence composition-based methods.

Discussions

HGT is one of the three major reasons for phylogenetic tree discordance among genes in a genome or between a gene and the species [69]. Recognition of HGT as a potent force driving bacterial evolution shook the ‘Tree of Life’ metaphor to

describe the evolutionary histories of and relationships between all organisms on earth. Several studies have proposed a ‘Network of Life’ that accounts for such discordance to understand the relationships among organisms [4, 5, 70, 71]. The phenomenon of HGT, and particularly the acquisition of GIs, confers novel features or functions to the recipient organisms, resulting in their diversification and adaptation.

Due to the profound impact of GIs on prokaryotic genome evolution, tremendous efforts have been made for decades to develop accurate methods for their detection. However, the current state of the art in GI detection leaves much to be desired. The existing algorithms suffer severely from imbalance in precision–recall trade-off, with precise methods tending to miss many GIs, whereas sensitive methods being prone to generating unacceptably high number of false positives.

In this paper, we have attempted to advance the state of the art in GI detection by attending to the issues of both precision and recall. Through a comprehensive literature survey followed by experiments involving large datasets, we have identified the characteristics of laterally transferred genes that are recognizable from the raw sequence alone. To make the method adaptive and independent of genome annotation, we have developed an unsupervised machine learning-based pipeline. A meticulously designed postprocessing in SSG-LUGIA ensures robust delineation of GIs in prokaryotic genomes. The ability to work with raw genome sequence alone and the unsupervised nature of the algorithm make it suitable for application to diverse sets of genomes, ranging from the well-studied genomes to newly sequenced, yet uncharacterized genomes.

A unique feature of SSG-LUGIA is its ability to control the precision and recall by tuning the parameters based on the application necessity. Though the algorithm is biased toward attaining high recall, it can be readily tuned to be precise while maintaining a relatively high F1-score. Overall, SSG-LUGIA outperformed the existing methods on benchmark datasets. SSG-LUGIA was occasionally outperformed by GIHunter, which may be because of incorporation of additional information (e.g. marker genes) by GIHunter and usage of biased test data (e.g. prevalence of GIs enriched in marker genes). Notably, these methods have complementary strengths—SSG-LUGIA identified GIs that were missed by GIHunter and vice versa. This illustrates the usefulness of in-tandem use of these methods to more comprehensively catalog GIs in prokaryotic genomes. SSG-LUGIA has been released as an

open-source software to help researchers including systematists probe heterogeneous prokaryotic genomes for LGT events.

Software availability and implementation

The algorithmic pipeline of SSG-LUGIA is implemented in Python 3 programming language [72]. The *k*-mer frequencies are efficiently computed using a sliding window backed by *Deque* data structure. We use Scikit-learn library [55] for the implementation of PCA and Elliptic Envelope for anomaly detection. All the codes are available at <https://github.com/nibtehz/SSG-LUGIA>.

A user can input an unannotated genome sequence to the software and the probable GIs will be predicted and their coordinates provided in an output file. Furthermore, if the user provides the gene annotation along with genome sequence, a list of genes harbored by predicted GIs will also be provided. We provide the SSG-LUGIA_P, SSG-LUGIA_R and SSG-LUGIA_F models with the software, but the user can set parameters according to the application need.

Key Points

- We present a comprehensive assessment of a compendium of computational protocols to solve an important problem in the fields spanning microbiology, evolutionary biology and biomedicine.
- We present a novel method, SSG-LUGIA, for identifying Genomic Islands (GIs) and horizontally transferred genes. SSG-LUGIA is an unsupervised machine learning-based method that obviates the need of functional annotation for localizing GIs.
- We highlight the strengths and weaknesses of different computational tools for GI prediction and propose an approach tailored to the choice of users (high sensitivity, high precision or a good balance between sensitivity and precision).
- The ability of SSG-LUGIA to work with raw genome sequence alone and the unsupervised nature of the algorithm make it suitable for application to diverse sets of genomes, ranging from well-studied genomes to newly sequenced, yet uncharacterized genomes.

Data availability

The datasets underlying this article were derived from sources in the public domain:

- The GIs for the 118 genomes used in IslandPick benchmark has been collected from the Supplementary Materials of [14], the corresponding sequences were collected from Entrez, molecular biology database, using biopython apis.
- The GIs and the sequences for the three genomes used in the GI-SVM [21] were collected from their official github repository https://github.com/icelu/GI_Prediction/tree/master/GI_SVM.

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